

Project Description: Intelligent Multimodal Search Engine

Group Number: 20

Client: Bernard Cheah Jun Kai

Company: Wysetime

This project focuses on building an intelligent multimodal search engine that enables users to search for images using natural language queries, addressing the limitations of traditional keyword-based systems. By leveraging a Vision-Language Model (VLM) such as OpenAI's CLIP, the system will interpret and encode both text and image data into a shared semantic space. This allows it to retrieve images based on meaning and context rather than manual tags or metadata.

The image data used in this project will be publicly sourced from online datasets such as Unsplash Lite, COCO, or Flickr30k, ensuring that all content is open and ethically accessible for research and prototyping purposes.

A LanceDB vector database will serve as the storage and retrieval backbone, optimized for high-performance similarity searches across millions of image embeddings. Its serverless design enables efficient querying, fast indexing, and seamless integration into Python-based machine learning workflows.

The project will progress through five structured phases:

1. Data Acquisition & Setup – sourcing and preparing public datasets.
2. Embedding Generation – using the VLM to create multimodal embeddings for images and text.

3. Database Indexing – storing embeddings in LanceDB and optimizing for vector similarity search.
4. Search Engine Development – building the core retrieval logic and a web-based interface (Streamlit/Flask).
5. Evaluation & Documentation – assessing retrieval accuracy (Precision@K), latency, and usability.

The final deliverable will be a functional prototype demonstrating context-aware image retrieval through natural language queries such as *"a dog catching a frisbee in a sunny park."* Additional outputs include a documented codebase, project report, performance metrics, and a presentation showcasing the system in action.

By integrating deep learning and vector database technologies, this project demonstrates how semantic understanding can transform image search into an intuitive, human-like experience.